



MASALALAR YECHISH

33- mavzu

1 Massasi 100 g bo'lgan metall jism 20 °C dan 130 °C ga cha isishi uchun 4180 J issiqlik miqdori olgan. Shu metall jismning solishtirma issiqlik sig'imini aniqlang.

Berilgan:	Formula	Hisoblash
$m = 0,1 \text{ kg}$ $Q = 4180 \text{ J}$ $t_1 = 20 \text{ }^{\circ}\text{C}$ $t_2 = 130 \text{ }^{\circ}\text{C}$	$Q = mc(t_2 - t_1)$ $c = \frac{Q}{m(t_2 - t_1)}$	$c = \frac{4180}{0,1 \cdot (130 - 20)} \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}} = 380 \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}}$
Topish kerak: $c = ?$	$[c] = \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}}$	Javob: $c = 380 \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}}$

2 Hajmi 20 l bo'lgan idishda 45 °C temperaturadagi suv, ikkinchi 30 l hajmli idishda esa 20 °C temperaturali suv bor. Har bir idishdagi suvda qanday issiqlik miqdori bor?

Berilgan:	Formula	Hisoblash
$V_1 = 20 \text{ l} = 20 \cdot 10^{-3} \text{ m}^3$ $V_2 = 30 \text{ l} = 30 \cdot 10^{-3} \text{ m}^3$ $t_1 = 45 \text{ }^{\circ}\text{C}$ $t_2 = 20 \text{ }^{\circ}\text{C}$ $c = 4200 \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}}$ $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$	$m = \rho \cdot V$ $Q = mc(t_2 - t_1)$ $(t_2 - t_1) = \frac{Q}{mc}$ $[m] = \frac{\text{kg}}{\text{m}^3} \cdot \text{m}^3 = \text{kg}$ $[Q] = \text{kg} \cdot \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}} \cdot ^{\circ}\text{C} = \text{J}$	Suv massasini uning hajmi orqali ifodalab hisoblaymiz: $m_1 = \rho V_1 = 1000 \cdot 20 \cdot 10^{-3} \text{ kg} = 20 \text{ kg}$ $m_2 = \rho V_2 = 1000 \cdot 30 \cdot 10^{-3} \text{ kg} = 30 \text{ kg}$ Temperaturasi t_1 bo'lgan suvdagi issiqlik miqdori: $Q_1 = m_1 c t_1 = 20 \cdot 4200 \cdot 45 \text{ J} = 3780000 \text{ J}$ Temperaturasi t_2 bo'lgan suvdagi issiqlik miqdori: $Q_2 = m_2 c t_2 = 30 \cdot 4200 \cdot 20 \text{ J} = 2520000 \text{ J}$
Topish kerak: $Q_1 = ?$ $Q_2 = ?$	$[t] = \frac{Q}{mc} = \frac{\text{J}}{\text{kg} \cdot \frac{\text{J}}{\text{kg} \cdot ^{\circ}\text{C}}} = ^{\circ}\text{C}$	Javob: birinchi idishdagi suvning issiqlik miqdori ko'p.

3 Hajmi 4 l bo'lgan idishda temperaturasi 100 °C bo'lgan qaynoq suv bor. Ma'lum vaqt o'tgach, idishdagi suv atrofga o'zidagi 672 kJ issiqlik miqdorini uzatib sovigan bo'lsa, suvning temperaturasi qanday bo'lib qolgan?

Berilgan:	Formula	Hisoblash
$V = 4 \text{ l} = 4 \cdot 10^{-3} \text{ m}^3$ $t_1 = 100 \text{ }^\circ\text{C}$ $Q_b = 672000 \text{ J}$ $c = 4200 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}$ $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$	$m = \rho \cdot V$ $Q = mc(t_2 - t_1)$ $(t_2 - t_1) = \frac{Q}{mc}$ $[m] = \frac{\text{kg}}{\text{m}^3} \cdot \text{m}^3 = \text{kg}$ $[Q] = \text{kg} \cdot \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}} \cdot ^\circ\text{C} = \text{J}$ $[t] = \frac{Q}{mc} = \frac{\text{J}}{\text{kg} \cdot \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}} = ^\circ\text{C}$	Suv massasini uning hajmi orqali ifodalab hisoblaymiz: $m_1 = \rho V_1 = 1000 \cdot 4 \cdot 10^{-3} \text{ kg} = 4 \text{ kg}$ Temperaturasi t_1 bo'lgan qaynoq suvdagi issiqlik miqdori: $Q_1 = m_1 c t_1 = 4 \cdot 4200 \cdot 100 = 1680000 \text{ J}$ Atrofga Q_b issiqlik miqdori uzatgandan so'ng suvda qolgan issiqlik miqdori: $Q = Q_1 - Q_b = 1680000 \text{ J} - 672000 \text{ J} = 1008000 \text{ J}$ Issiqlik miqdori formulasiga ko'ra: $Q = m c t$ Bundan $t = \frac{Q}{m c} = \frac{1008000}{4 \cdot 4200} = 60^\circ\text{C}$ Javob: $t_2 = 60^\circ\text{C}$.
Topish kerak: $t_2 = ?$		



22-mashq

1 Massasi 200 g bo'lgan alyuminiydan yasalgan jismni 20°C dan 300°C gacha qizdirish uchun qancha issiqlik miqdori kerak bo'ladi?

2 Misdan yasalgan detal temperaturasi 420°C dan 40°C gacha sovidi. Agar uning hajmi 200 cm^3 bo'lsa, undan qancha issiqlik miqdori ajralgan?

3 Massasi 3 kg bo'lgan jism 20°C dan 500°C gacha qiziganda $1267,2 \text{ kJ}$ issiqlik miqdori olgan bo'lsa, bu jism qanday moddadan tayyorlangan?

4 Massasi 2,5 kg bo'lgan jism temperaturasi 28°C dan 100°C ga ko'tarilguncha $68,4 \text{ kJ}$ issiqlik miqdorini olgan. Jism yasalgan modda turini aniqlang.

5 Eni 3 m, uzunligi 5 m va chuqurligi 2 m bo'lgan hovuzdagi suvni 17°C dan 30°C gacha isitish uchun qancha issiqlik miqdori kerak bo'ladi?

6 Normal sharoitda temperaturasi 20°C va hajmi 1,5 l bo'lgan suv qaynaguncha qancha issiqlik miqdorini oladi?

7 Grafikda massasi 200 g bo'lgan jismga issiqlik miqdori berilganda uning temperaturasi o'zgarishi keltirilgan. Grafikdan foydalanib jismning turini aniqlang.

8 Hajmi 2 l bo'lgan idishda temperaturasi 100°C bo'lgan qaynoq suv bor. Ma'lum vaqt o'tgach, idishdagi suv atrofga o'zidagi issiqlik miqdorining 40 foizini uzatib sovigan bo'lsa, suvning temperaturasi qanday bo'lib qolgan?

